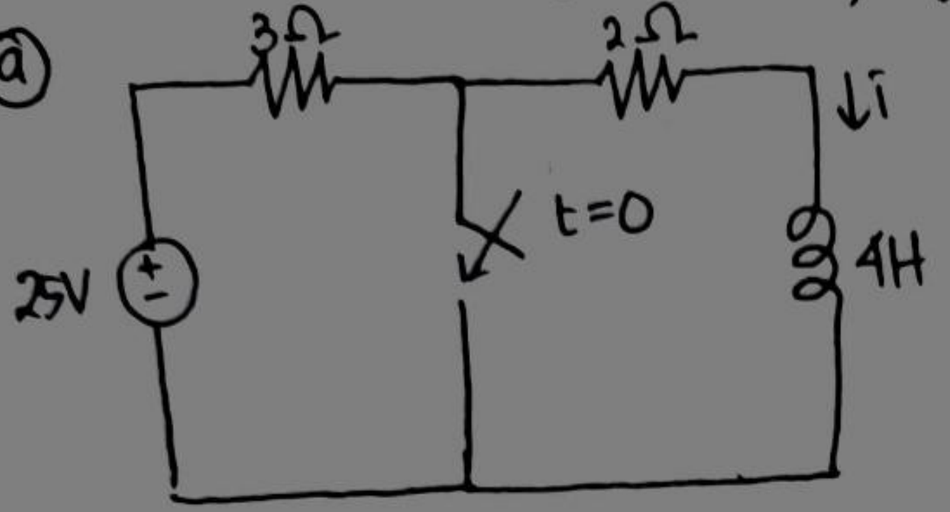
Dinda Qurrota Aini Al-sefy
20133103 8
1B-TTE

1. Tentukan nilai arus (i) dan tegangan (v) di sisi masing2 induktor, untuk $t>0$.

A. ⓐ



jawab:

- saat $t<0$, saklar terbuka,

$$I_0 = \frac{V}{R} = \frac{25}{3+2} = 5A.$$

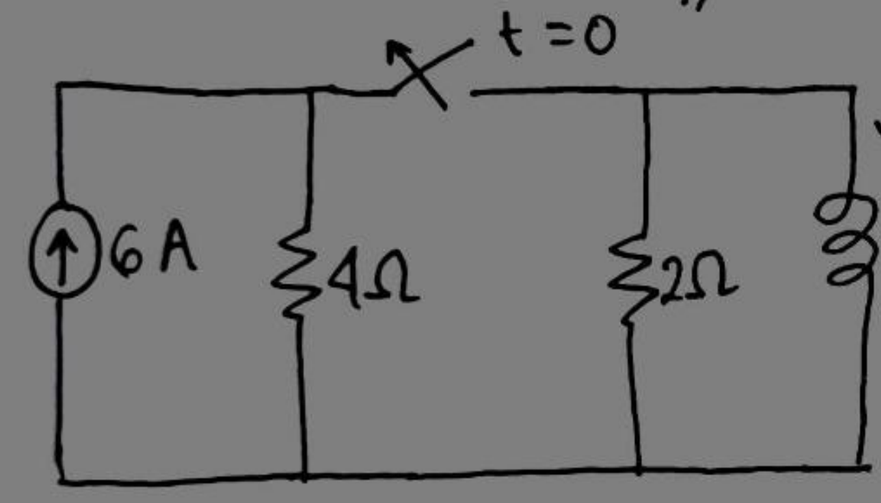
- saat $t>0$, saklar tertutup, induktor aktif

$$\tau = \frac{L}{R} = \frac{4}{2} = 2s$$

$$i(t) = I_0 e^{-t/\tau}$$
$$= 5e^{-t/2} A$$
$$= 5e^{-0,5t} A //$$

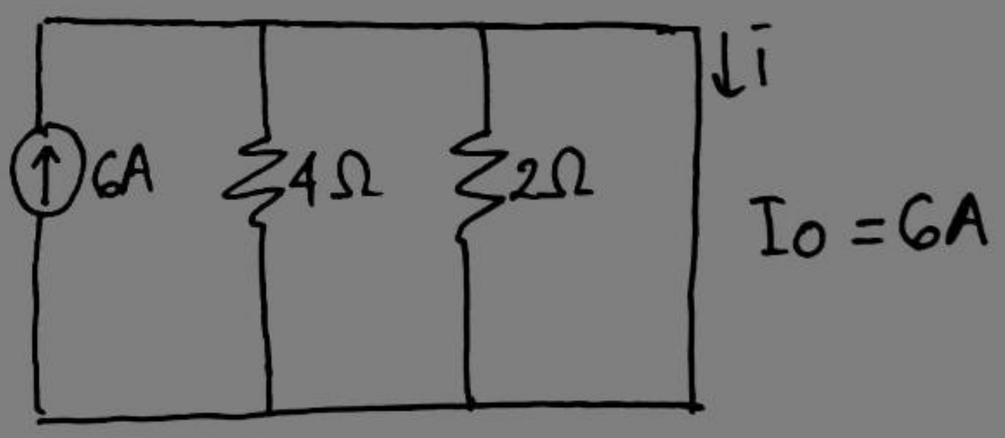
$$V = L\frac{di}{dt}$$
$$= 4(-2,5e^{-0,5t} A)$$
$$= -10e^{-0,5t} V //$$

ⓑ



jawab:

- saat $t<0$, saklar tertutup, induktor short circuit



$I_0 = 6A$

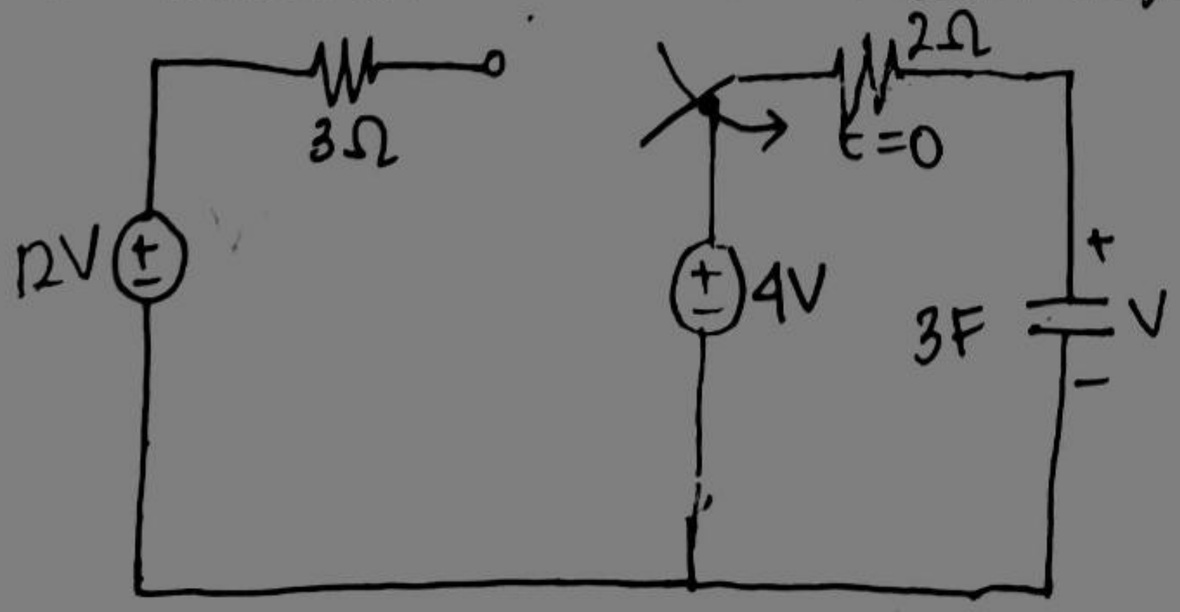
- saat $t>0$, saklar terbuka, induktor aktif

$$\tau = \frac{L}{R} = \frac{3}{2} = 1,5s$$

$$i(t) = I_0 e^{-t/\tau}$$
$$= 6e^{-t/1,5s} A$$
$$= 6e^{-2/3t} A //$$

$$v = L\frac{di}{dt}$$
$$= \cancel{3}\left(6 \cdot -\frac{2}{\cancel{3}} e^{-2/3t} A\right)$$
$$= -12e^{-2/3t} V //$$

B. Tentukan besar arus (i) dan tegangan (v) dikapasitor



Jawab

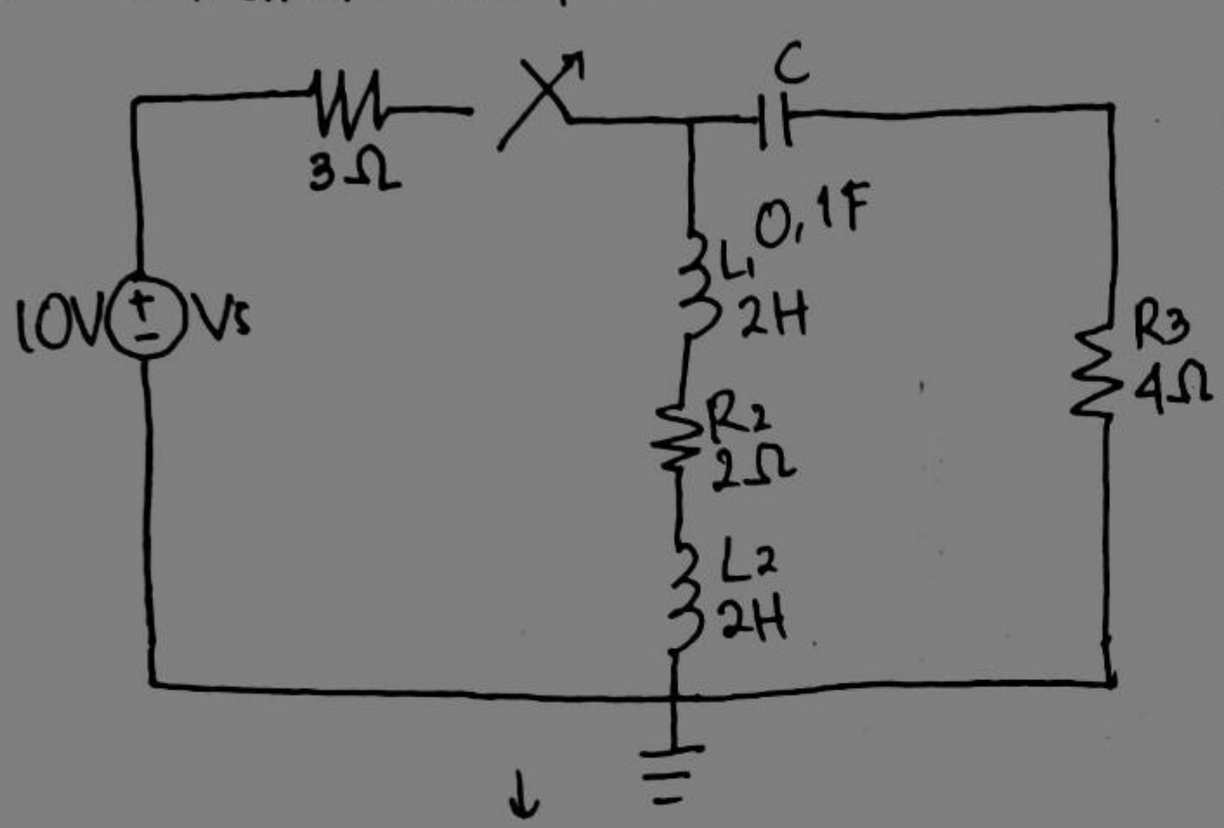
- saat $t<0$, C sebagai open circuit,
  $V(t) = 12V$
- saat $t>0$, maka
  $\tau = RC = 2 \times 3 = 6s$
  $V(t) = V_s + (V_0 - V_s)e^{-t/\tau}$

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$$V(t) = 4 + (12-4)e^{-t/6}$$
$$V(t) = 4 + 8e^{-t/6}\ V$$

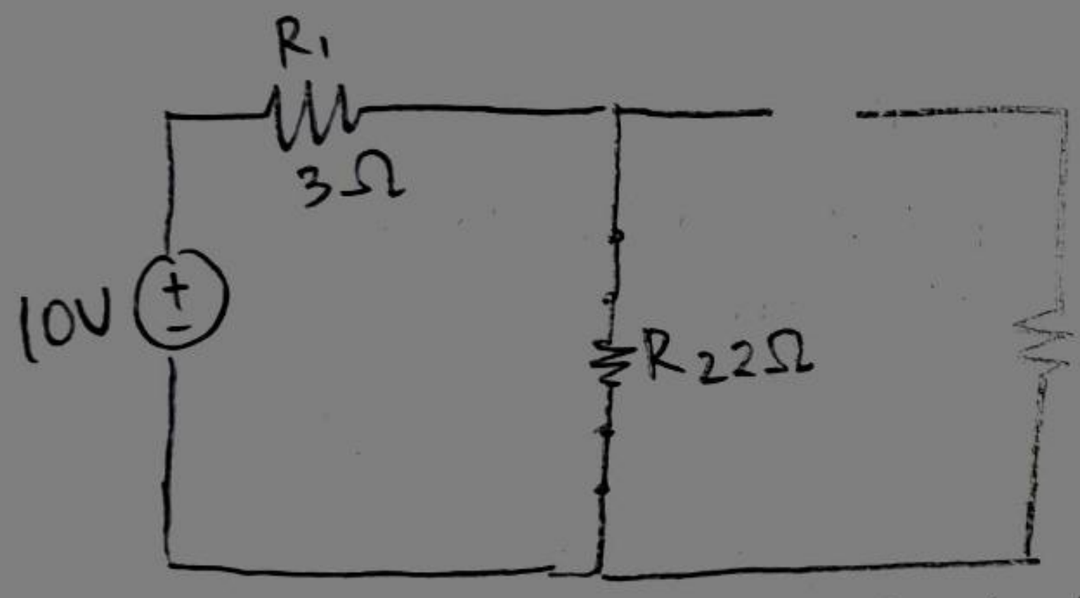
$$i = C\frac{dv}{dt} = \frac{1}{10}\left(-\frac{1}{6}\cdot 8e^{-t/6}\right)$$
$$= -4e^{-t/6}\ A.$$

② Tentukan berapa besar nilai v(t) dan tegangan di R_2, $t>0$



Jawab.

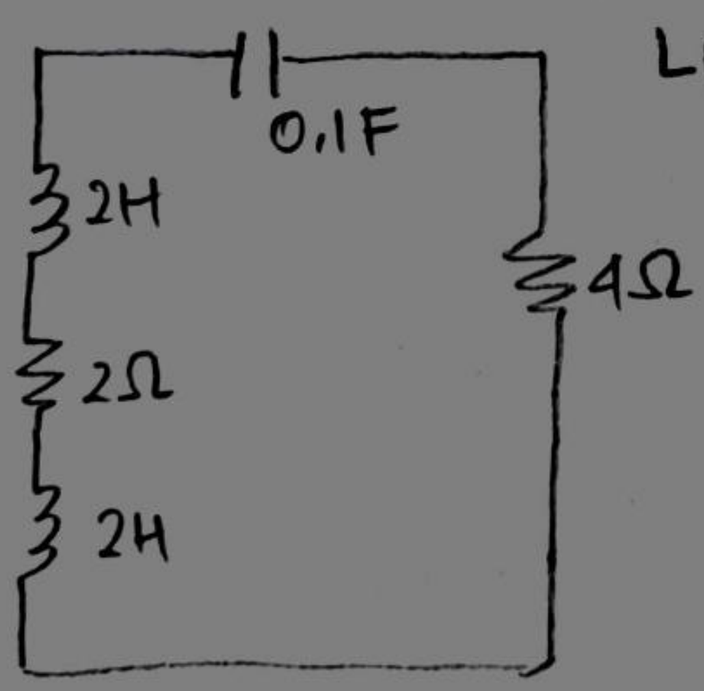
- saat $t \leq 0$, saklar tertutup, kapasitor open circuit, L hubung singkat



$V(t) = V_{2\Omega} = 10 \times \frac{2}{3+2} = 10 \times \frac{2}{5}$
$= 4V$

$V(0) = 4V$

- saat $t > 0$, saklar terbuka.



$L_P = 2+2 = 4H$

$\alpha = \frac{2+4}{2 \cdot 4} = \frac{6}{8} = 0,75$

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{4 \cdot 0,1}} = 1,58\ rad/s$

$\alpha < \omega_0 \rightarrow$ underdamped

$V(t) = (A_1 \cos \omega t + A_2 \sin \omega t)e^{-\alpha t}$

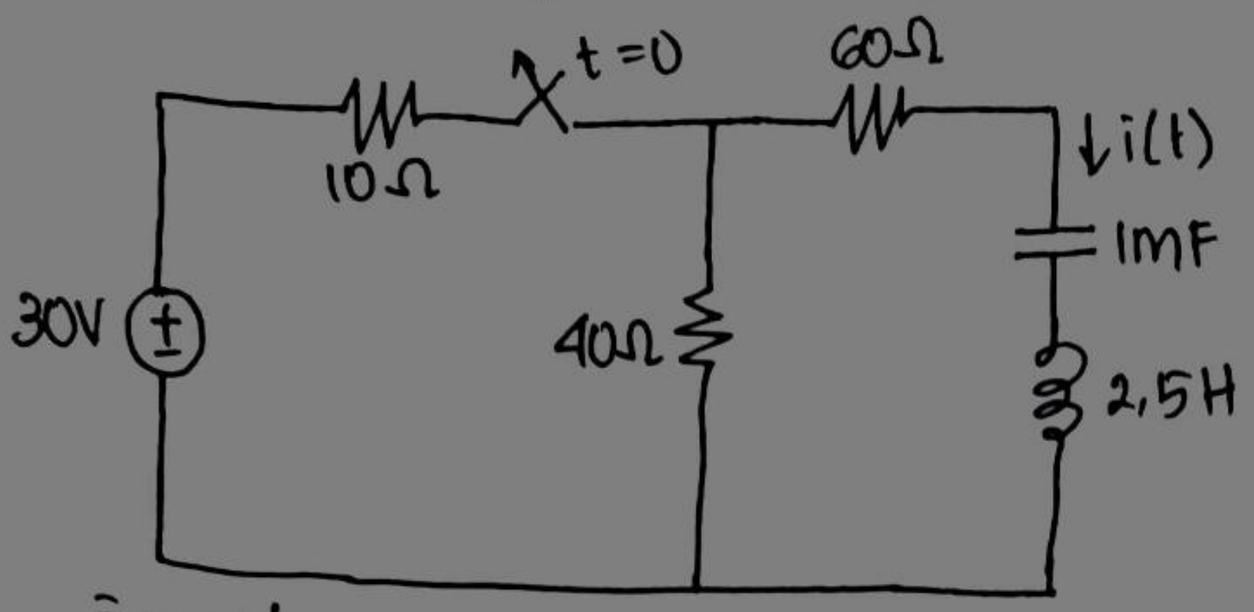
$t = 0$

$4V = (A_1 + 0)e^{-0}$

$4V = A_1$.

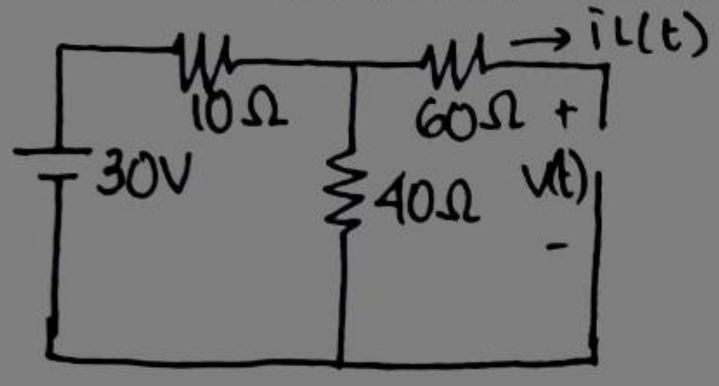
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③ Tentukan berapa besar nilai arus (t) dan tegangan di sisi ~~kapasitor~~, induktor



Jawab.

- Saat $t \leq 0$, saklar tertutup, capasitor open circuit, induktor hubung singkat



$$V(t) = V_{40\Omega} = 30 \cdot \frac{40}{40+10}$$

$$= \frac{6}{30} \times \frac{40}{50} = 24V, \quad V(0) = 24V$$

$i_L(t)$ saat $t \leq 0 = 0$, maka $i_L(0) = 0A$

- saat $t > 0$, saklar terbuka,



$$\alpha = \frac{R}{2L} = \frac{40+60}{2 \cdot 2{,}5} = \frac{100}{5} = 20 \text{ rad/s}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{2{,}5 \times 10^{-3}}} = 20 \text{ rad/s}$$

$\alpha = \omega_0 \rightarrow$ critically damped, maka

$v(t) = (A_2 + A_1 t)\, e^{-\alpha t} \rightarrow t > 0 \ldots\ldots$ ①

$t = 0$

$V(0) = (A_2 + 0)e^{-0}$

$= A_2 e^{-0}$

$V(0) = A_2$

$24V = A_2$

Turunkan persamaan 1

$$\frac{dv(t)}{dt} = A_1 e^{-\alpha t} - \alpha e^{-\alpha t}(A_2 + A_1 t)$$

$t = 0$

$$\frac{dv(0)}{dt} = A_1 - \alpha A_2 \ldots ②$$

→ RLC rangkaian seri

$i_L(t) = i_C(t)$

$i_L(0) = i_C(0) = 0$

$$i_C(0) = C \frac{dv(0)}{dt} \rightarrow \frac{dv(0)}{dt} = \frac{1}{C} i_C(0) \ldots ③$$

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maka

$$A_1 - \alpha A_2 = \frac{1}{C} i_C$$

$$A_1 - (20 \times 24) = \frac{1}{C} i_C(0)$$

$$A_1 = 480V$$

- $v(t) = (24 + 480t)e^{-20t} V, \quad t > 0$

$$i(t) = C \frac{dv(t)}{dt}$$

$$i(t) = 10^{-3} \times \frac{d}{dt}((24+480)t^{-20t} V)$$

$$= 10^{-3}(480e^{-20t} - 480e^{-20t} - 9600te^{-20t})$$

$$= 10^{-3}(-9600\, te^{-20t})$$

$$= -9{,}6\, te^{-20t} A //$$

$$V = L \frac{di}{dt} = 2{,}5 \left(\frac{-12 + 960t}{5-2}\right)e^{-20t}$$

$$= (-24 + 480t)e^{-20t}$$

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4. A. Tentukan nilai I_4, jika



i. $\bar{I}_1 = 6\cos 3t\,A$, $\bar{I}_2 = 4\cos(3t-30°)A$ dan $\bar{I}_3 = -4\sqrt{3}\cos(3t+60°)A$

Jawab.

$\bar{I}_1 = 6\cos 3t = 6\angle 0°$

$\bar{I}_2 = 4\cos(3t-30°) = 4\angle 30°$

$\bar{I}_3 = -4\sqrt{3}\cos(3t+60°) = -4\sqrt{3}\angle 60°$

KCL.

$-\bar{I}_1 - \bar{I}_2 - \bar{I}_3 + \bar{I}_4 = 0$

$\bar{I}_4 = \bar{I}_1 + \bar{I}_2 + \bar{I}_3$

$= 6\angle 0° + 4\angle 30° + 4\sqrt{3}60°$

$I_4 = 10\angle -53,13°\,A.$

$\bar{I}_4 = 10\cos(3t-53,13°)A$

ii. $\bar{I}_1 = 5\cos(3t+30°)A$, $\bar{I}_2 = 5\sin 3t\,A$, $\bar{I}_3 = 5\cos(3t+150°)A$.

$\bar{I}_1 = 5\angle 30°$

$\bar{I}_2 = 5\sin 3t = 5\cos(90-3t) = 5\angle 90°$

$\bar{I}_3 = 5\cos(3t+150°) = 5\angle 150°$

$I_4 = I_1 + I_2 + I_3$

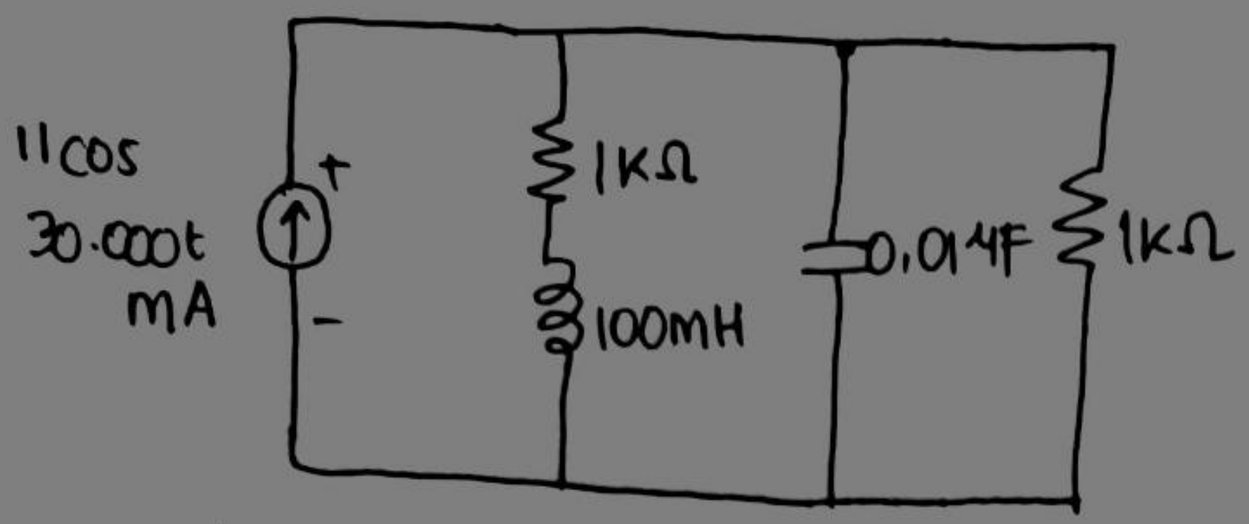
$= 5\angle 30° + 5\angle -90° + 5\angle 150°$

$= 15\angle 90°$

$= 15\cos(3t+90°)A$

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B. Tentukan besarnya V dan arus yang mengalir pada C=0,01μF



$\omega = 30 \times 10^3$

$i = 11 \cos 30.000t \text{ mA}$

$Im = 11 \text{ A}$

$X_L = \omega L$

$= 30 \times 10^{3} \cdot 100 \times 10^{-3}$

$= 3000\,\Omega$

$X_C = \frac{1}{\omega C} = \frac{1}{30 \times 10^3 \cdot 10^{-8}}$

$= \frac{1}{30 \times 10^{-5}} = \frac{10^5}{30} = 3333{,}3\,\Omega$

$R = 1k\Omega$

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